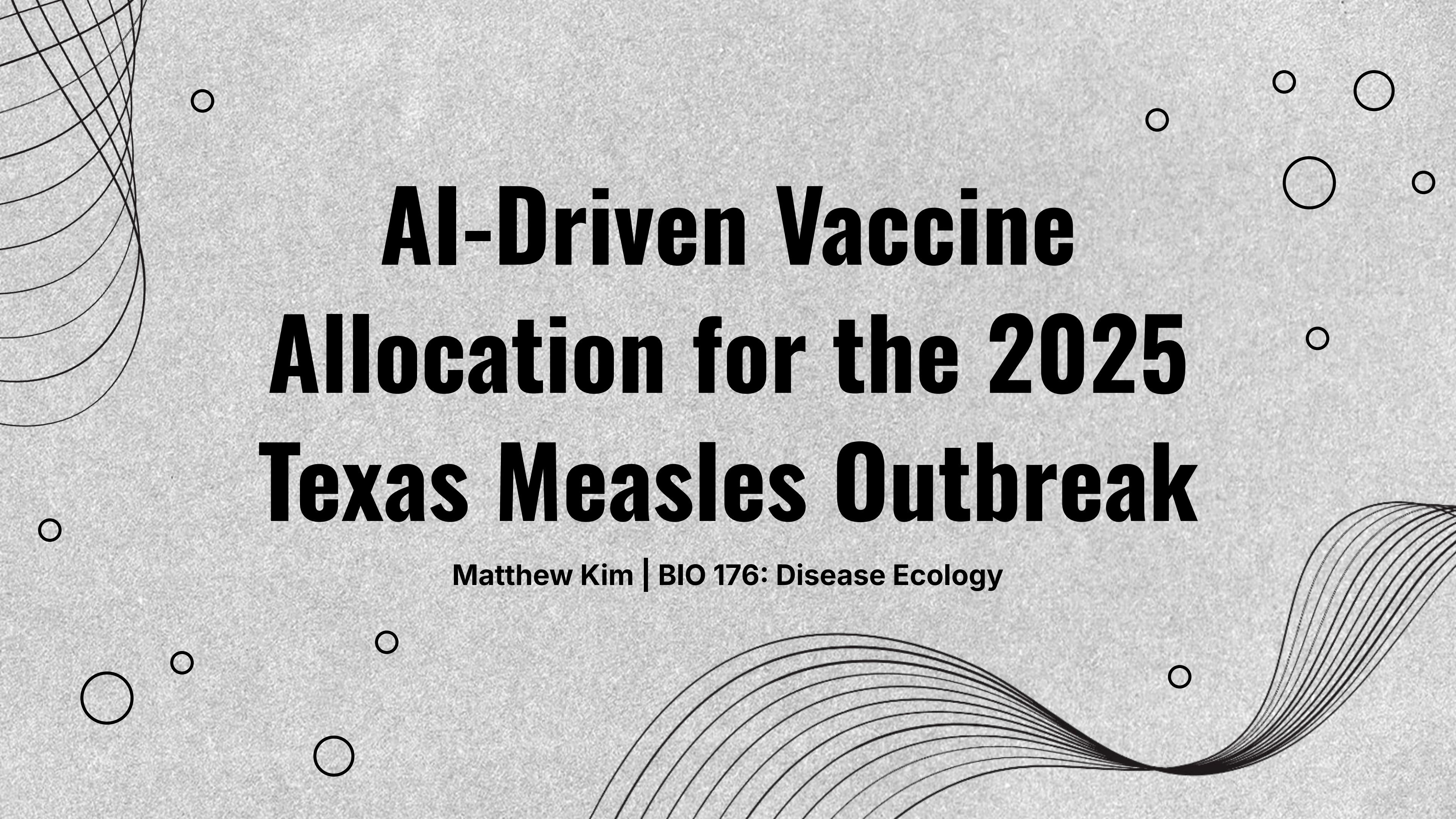




AI-Driven Vaccine Allocation for the 2025 Texas Measles Outbreak

Matthew Kim | BIO 176: Disease Ecology

Tracking U.S. Measles Outbreaks

By [Jonathan Corum](#) and [Teddy Rosenbluth](#) Updated Nov. 19, 2025



Share full article



407

The Centers for Disease Control and Prevention has reported [1,753 measles cases](#) as of Nov. 18, the highest number of annual cases

declared measles [eliminated](#) in 2000.

30 cases [TEXAS MEASLES OUTBREAK](#)

ly linke



Texas measles cases remain steady at 750. Track the spread here.

An outbreak is considered over if there are no new cases reported in 42 consecutive days.

SON

2.P.M. CENTRAL

REPUBLIC SHARE

Texas Measles Strain Continues to Spread, Officials Say

If the outbreaks cannot be extinguished soon, the United States will lose its “elimination status,” as determined by the World Health Organization.

Measles outbreak that sickened more than 700 declared over in Texas

The outbreak led to the deaths of two school-aged children.

By [Youri Benadjaoud](#)

August 18, 2025, 12:31 PM



How effective is AI-driven vaccine allocation in preventing the spread of measles?

texasmeaslesai.github.io

MATTHEW KIM | BIO 176: DISEASE ECOLOGY

Could AI Have Saved Texas From Measles?

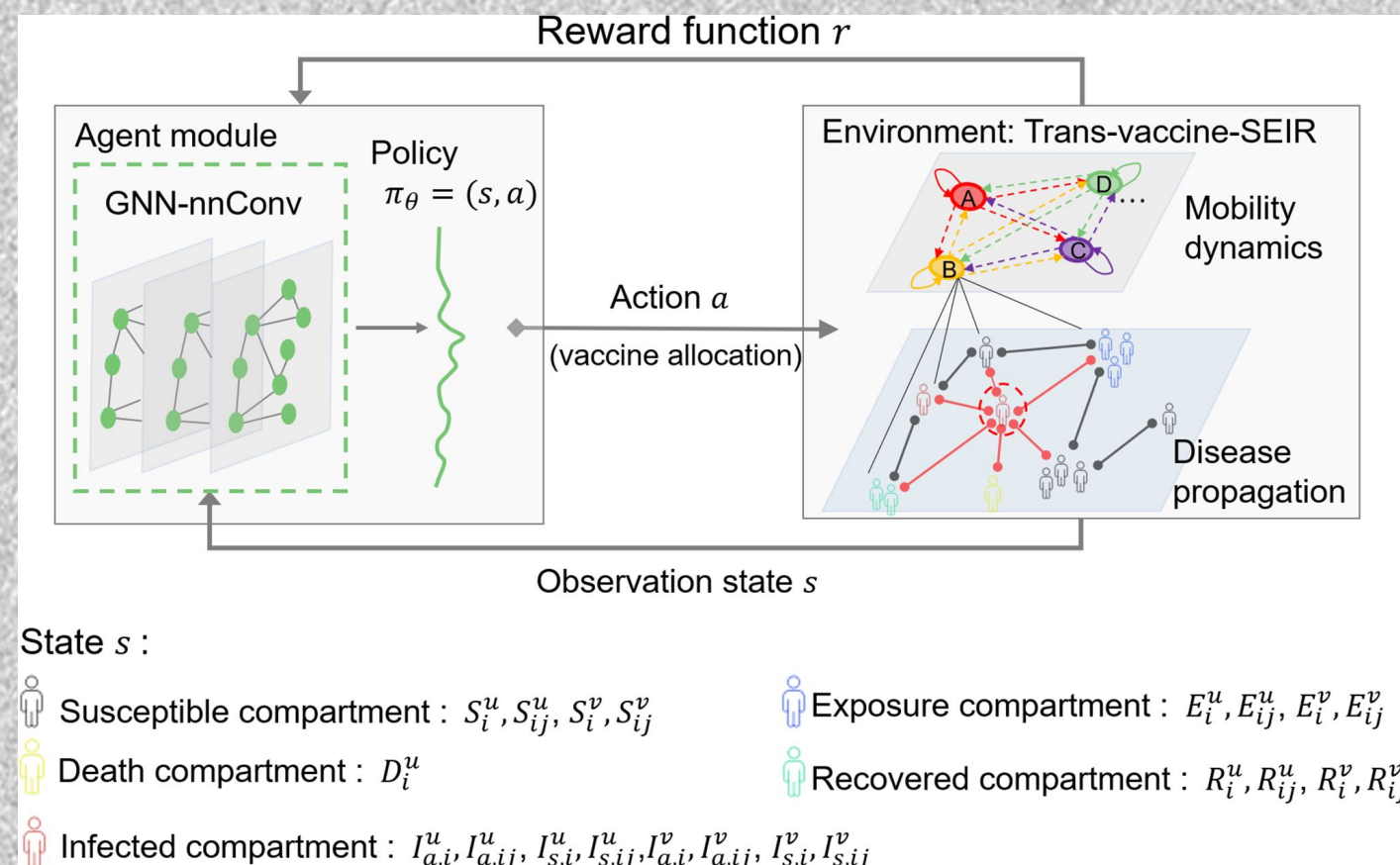
When a measles outbreak hits, every vaccine counts.

I tested if artificial intelligence could outsmart
standard heuristic vaccine allocation strategies.



Background Information

- Much prior work has been done with measles: R_0 between 12 and 18
- Frequency dependent transmission with nearly constant R_0
- Vaccination coverage plays a crucial role in the stability of measles transmission
- Reinforcement learning (RL) has been explored for vaccine allocation but not measles



Deep RL-based approach with mobility:
reduces 7%-10% of infections/deaths

ALGORITHM 1: PPO in Multi-City Vaccine Allocation

Input: $S_i(t), E_i(t), I_i(t), R_i(t), D_i(t)$ from the Environment for each city i

Output: decision $v(t) = [v_1(t), v_2(t), \dots, v_N(t)]$

for $t \in \{1, \dots, T\}$ **do**

 Actor network outputs decision $v(t)$

 Update Environment $S_i(t), E_i(t), I_i(t), R_i(t), D_i(t)$ from t to $t+1$

 Environment gives immediate reward $r_t = -\max(\sum_{i=1}^N (I_i(t+1) - I_i(t)), 0)$

 Update Critic network based on the immediate reward

 Critic network gives long-term reward $\mathbb{E} \left[\sum_{t=1}^T r_t \right]$

 Update Actor network based on the long-term reward

end

Proximal Policy Optimization (PPO) approach
with multi-city network for COVID-19:
reduces 8% of peak infections

Hypothesis

- The RL model can significantly reduce the spread of measles when few limitations exist (i.e., no vaccine hesitancy, high daily vaccine supply, etc.)
- When limitations become greater, the RL model will have more difficulties due to time needed to learn and adapt (not ideal for outbreaks)
- However, the RL model will still result in the lowest peak infections due to adaptability to changing factors compared to set heuristic methods

**Model
Fitting**

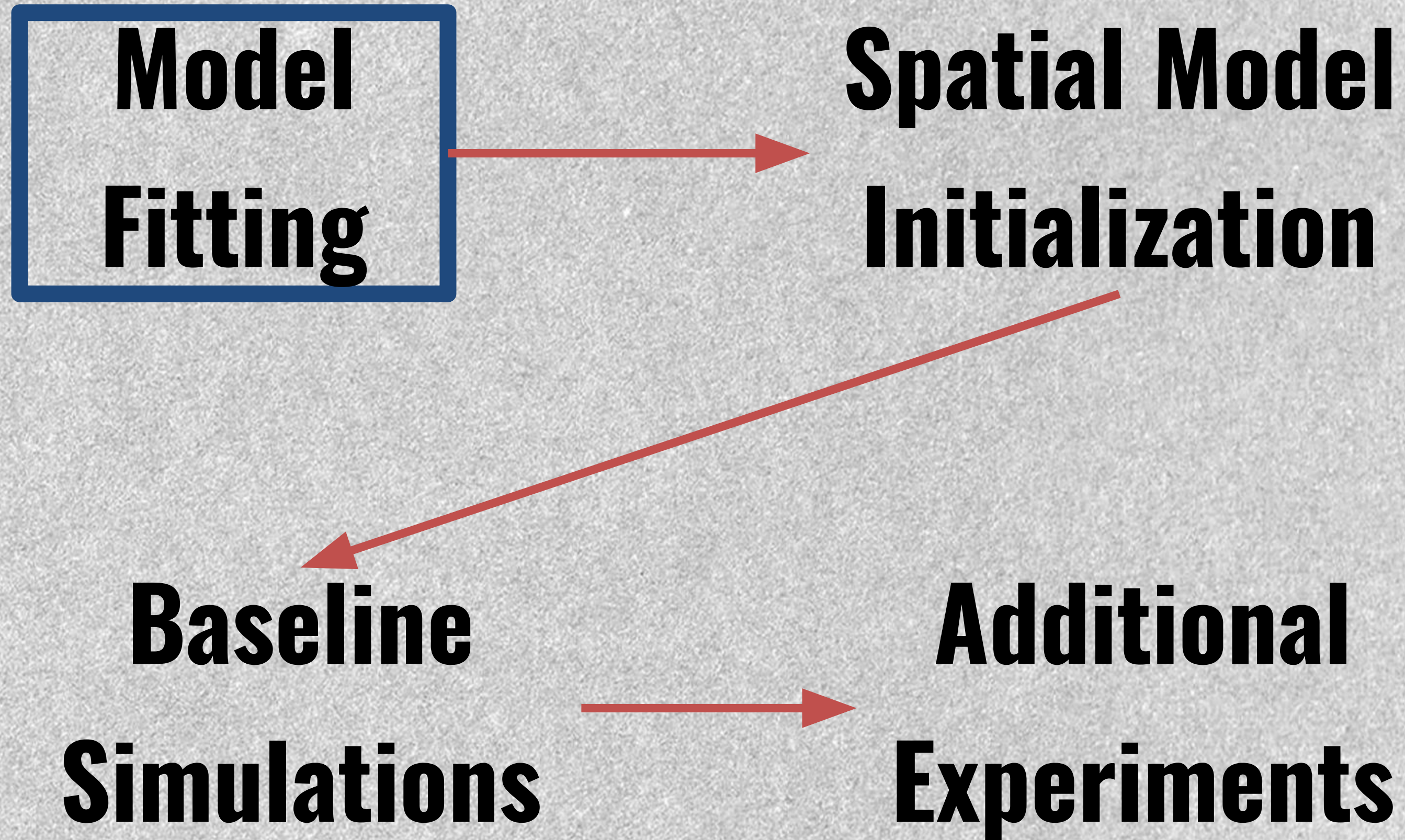
**Spatial Model
Initialization**



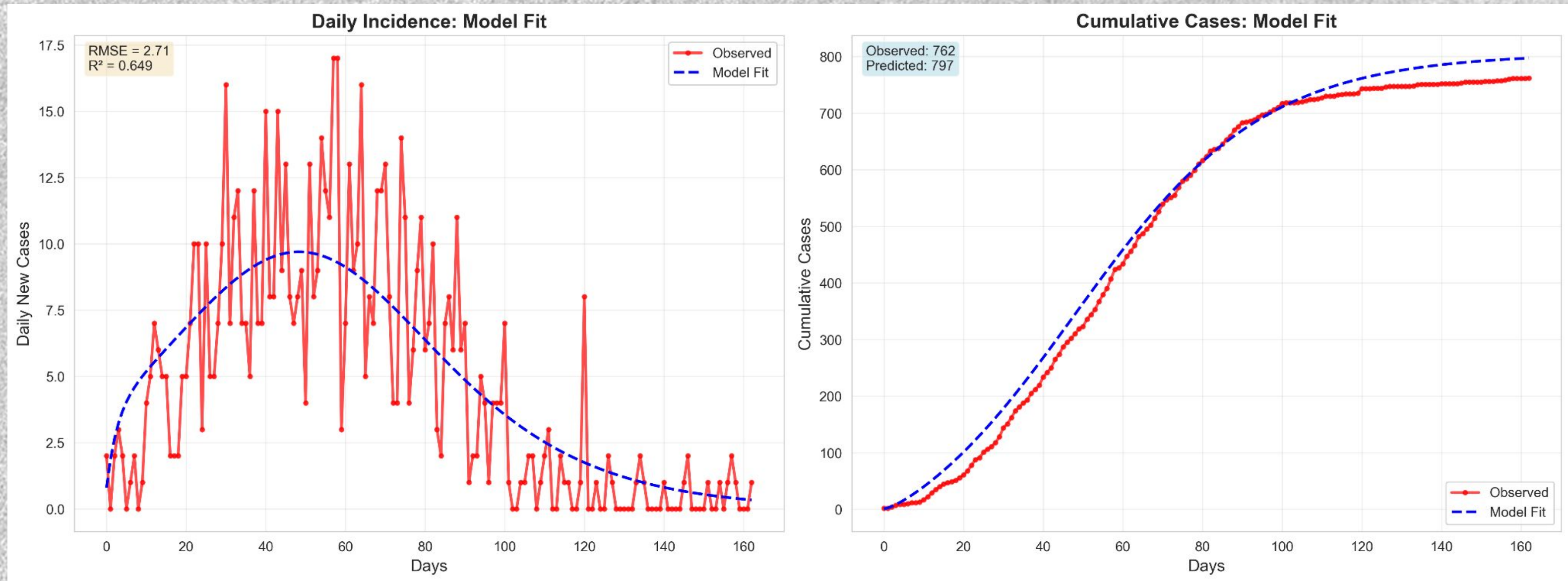
**Baseline
Simulations**

**Additional
Experiments**





Model Fitting



Initial fitting to single patch SEIR model:

$$dS/dt = -\beta(SI/N)$$

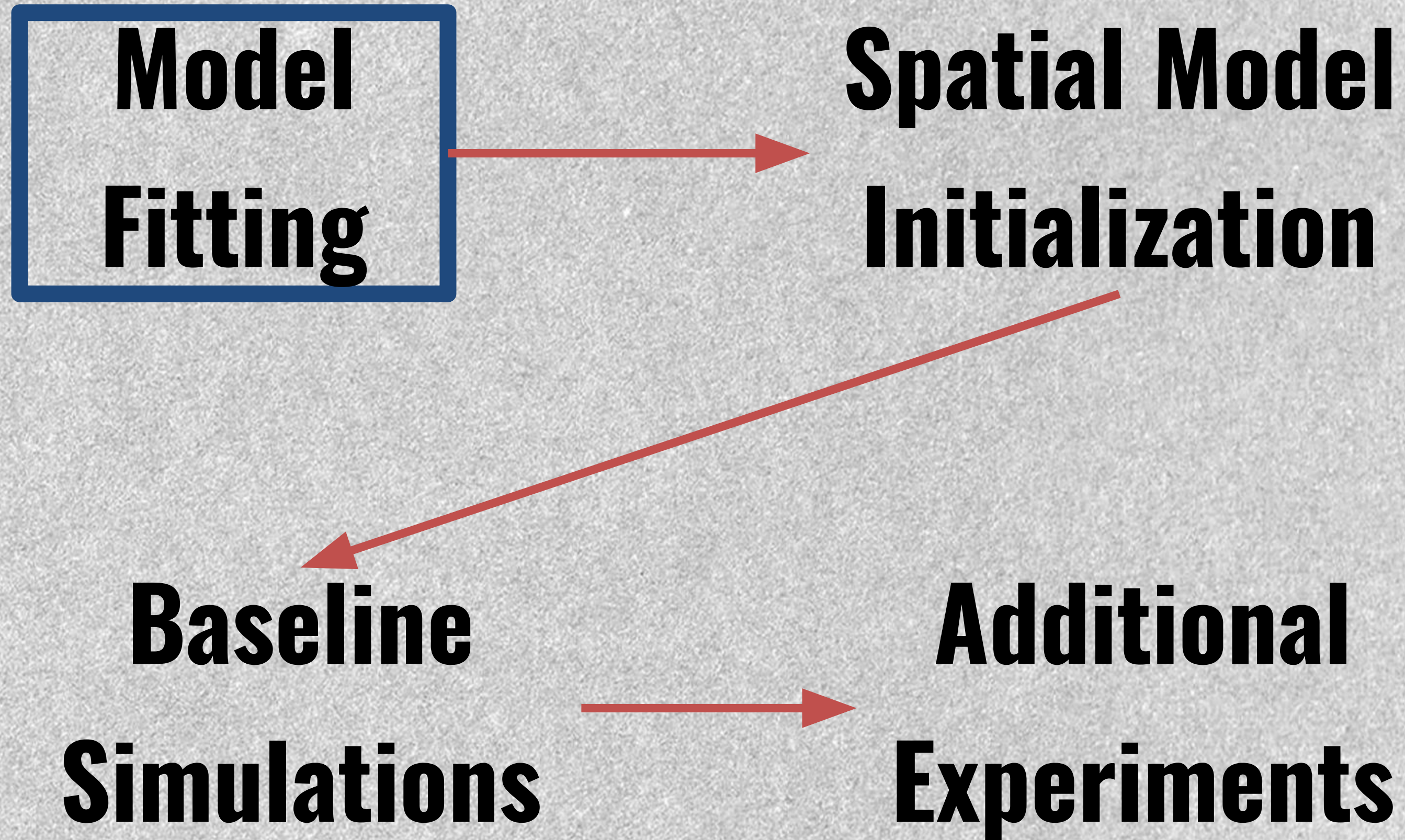
$$dE/dt = \beta(SI/N) - \sigma E$$

$$dI/dt = \sigma E - \gamma I$$

$$dR/dt = \gamma I$$

- Used Poisson deviance metric via the Nelder-Mead algorithm
- Set $N = 10000$, $\sigma = 1/8$, $\gamma = 1/7$

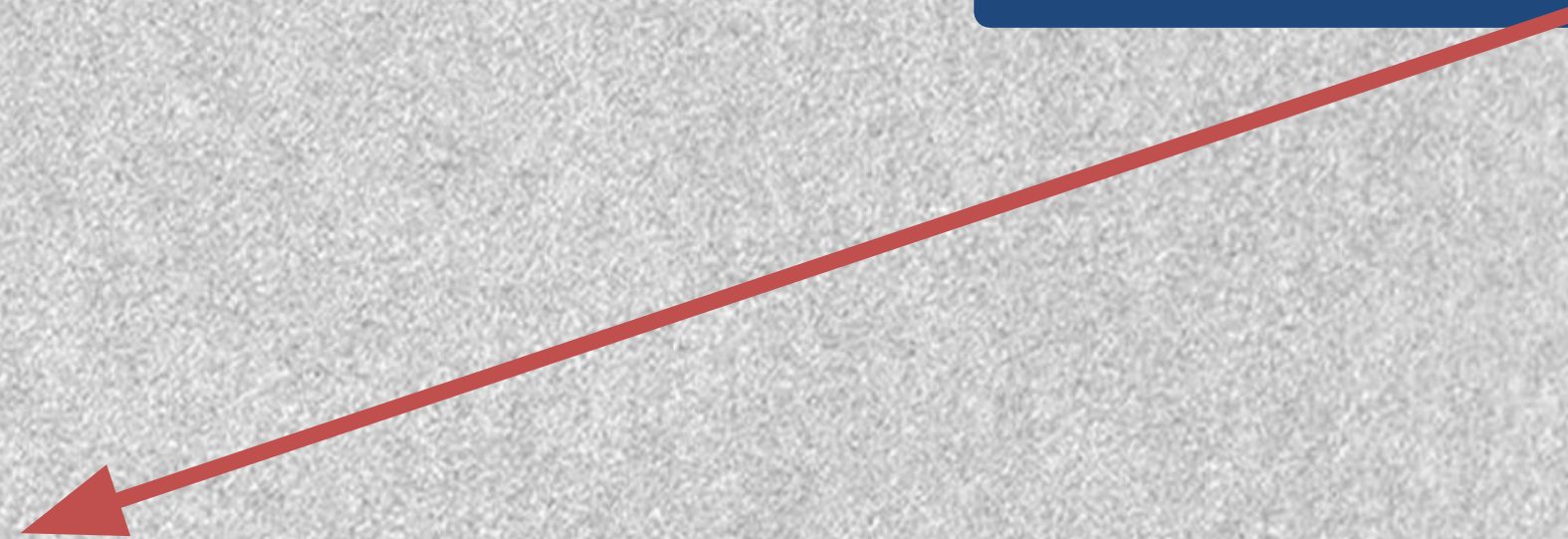
- $\beta = 2.000$
- $S_o = 1148$
- $E_o = 6$
- $I_o = 41$
- **$R_o = 14.00$**



**Model
Fitting**



**Spatial Model
Initialization**



**Baseline
Simulations**



**Additional
Experiments**

Multi-County SEIR Model

County Initialization

- Distributing the fitted parameters
- Example:
 - $S_{\text{county}} = S_{\text{fitted}} * (\text{Cases}_{\text{county}} / \text{Cases}_{\text{total}}) * 37$

Force of Infection (λ)

- Risk from local infections and infections in connected counties
- For county i:
 - $\lambda_i = \beta(I_i/N_i) + \sum_{j \neq i} m_{ij} \beta(I_j/N_j)$
 - where m_{ij} is the mobility coupling constant (set to 0.002 if cases > 10)

Vaccine Hesitancy

- $V_{\text{effective}} = V_{\text{allocated}} * (1 - H_i)$
- where H_i is the hesitancy rate fo county i

$$dS_i/dt = -\lambda_i(t) * S_i - v_i(t)$$

$$dE_i/dt = \lambda_i(t) * S_i - \sigma E_i$$

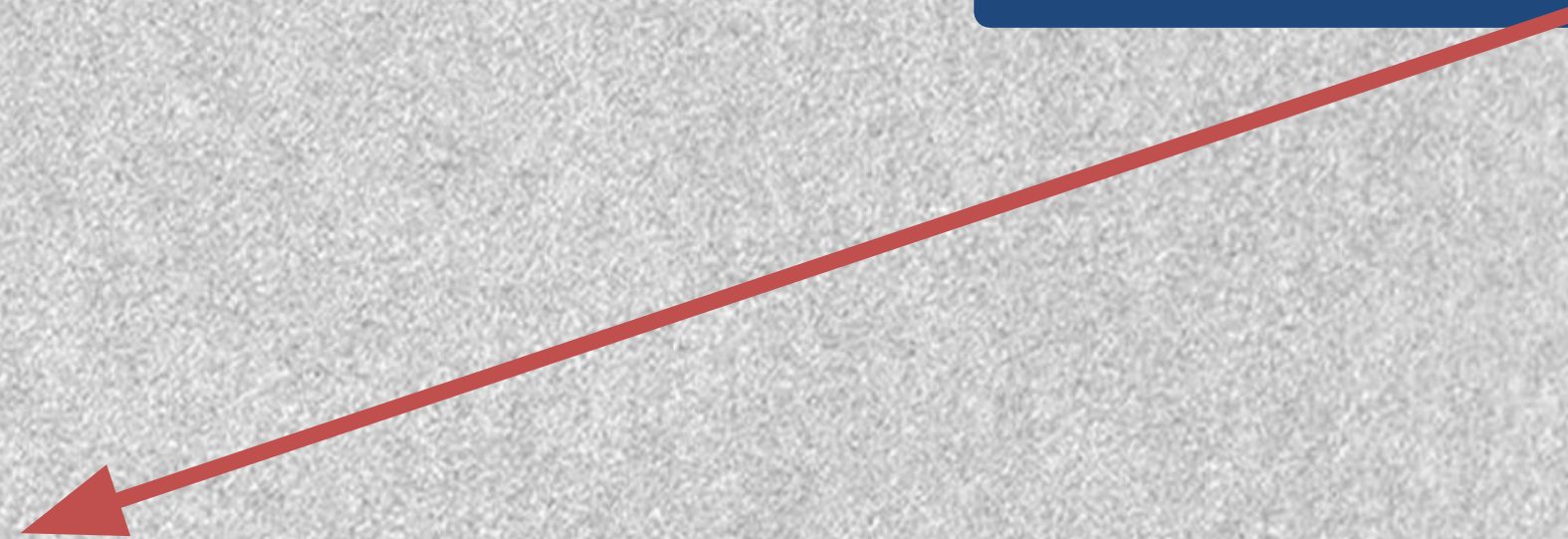
$$dI_i/dt = \sigma E_i - \gamma I_i$$

$$dR_i/dt = \gamma I_i + v_i(t)$$

**Model
Fitting**



**Spatial Model
Initialization**



**Baseline
Simulations**



**Additional
Experiments**

**Model
Fitting**

**Spatial Model
Initialization**

**Baseline
Simulations**

**Additional
Experiments**



Four Strategies

Reinforcement Learning Agent

- Q-Learning with ϵ -greedy exploration
- Defined by top-3 highest-risk counties and remaining vaccine budget status
- Reward function:
 - $R = (I_{\text{previous}} - I_{\text{current}}) * 100 - I_{\text{current}} + \text{BudgetBonus}$

Uniform Distribution

- Vaccines allocated equally across all counties

High-Risk Targeting

- Prioritizes allocation to counties with highest active infections
- Allocates up to 50% of susceptible population per county until supply exhausted

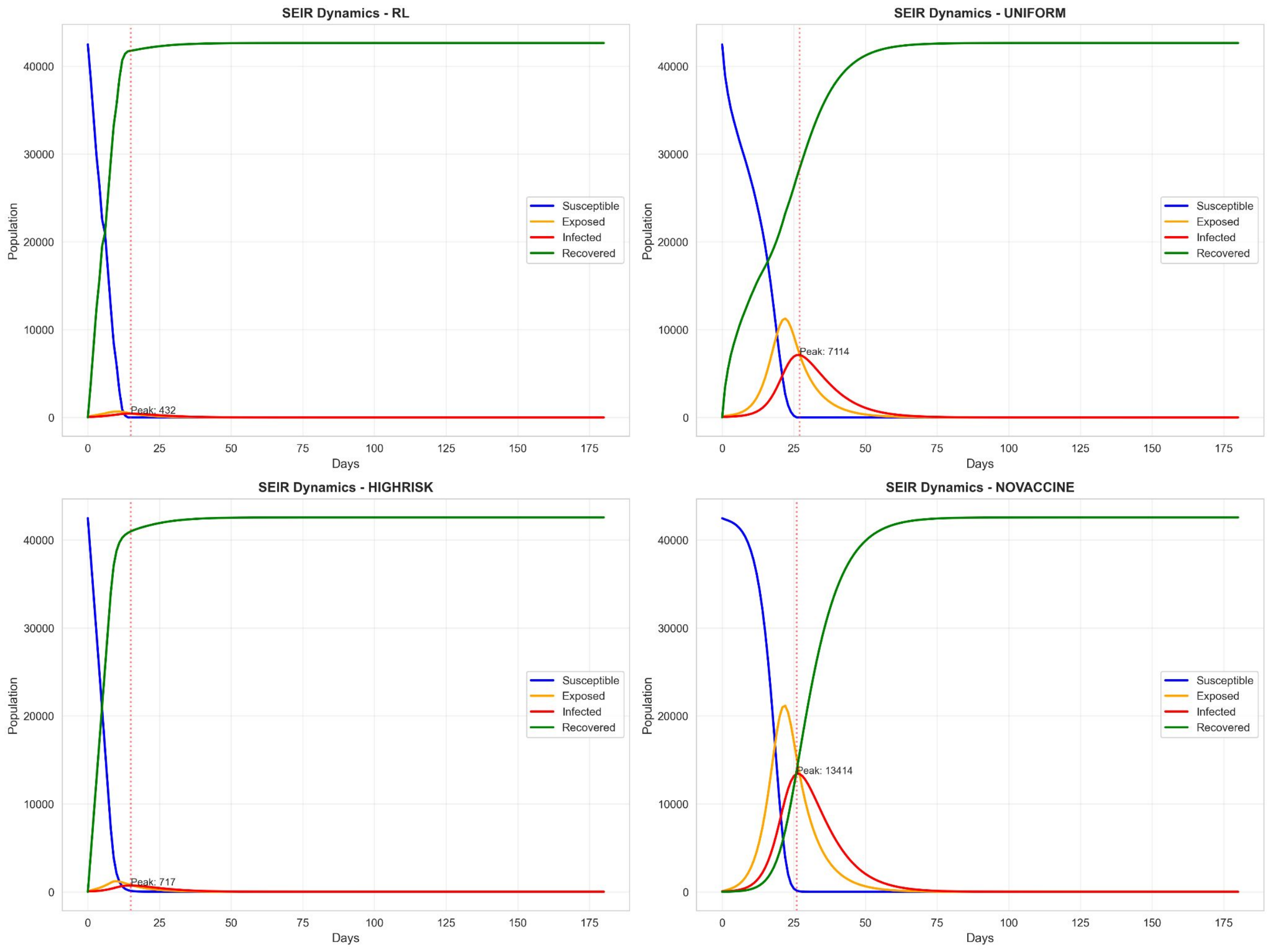
No Vaccination

- ￣_(\ツ)_/￣

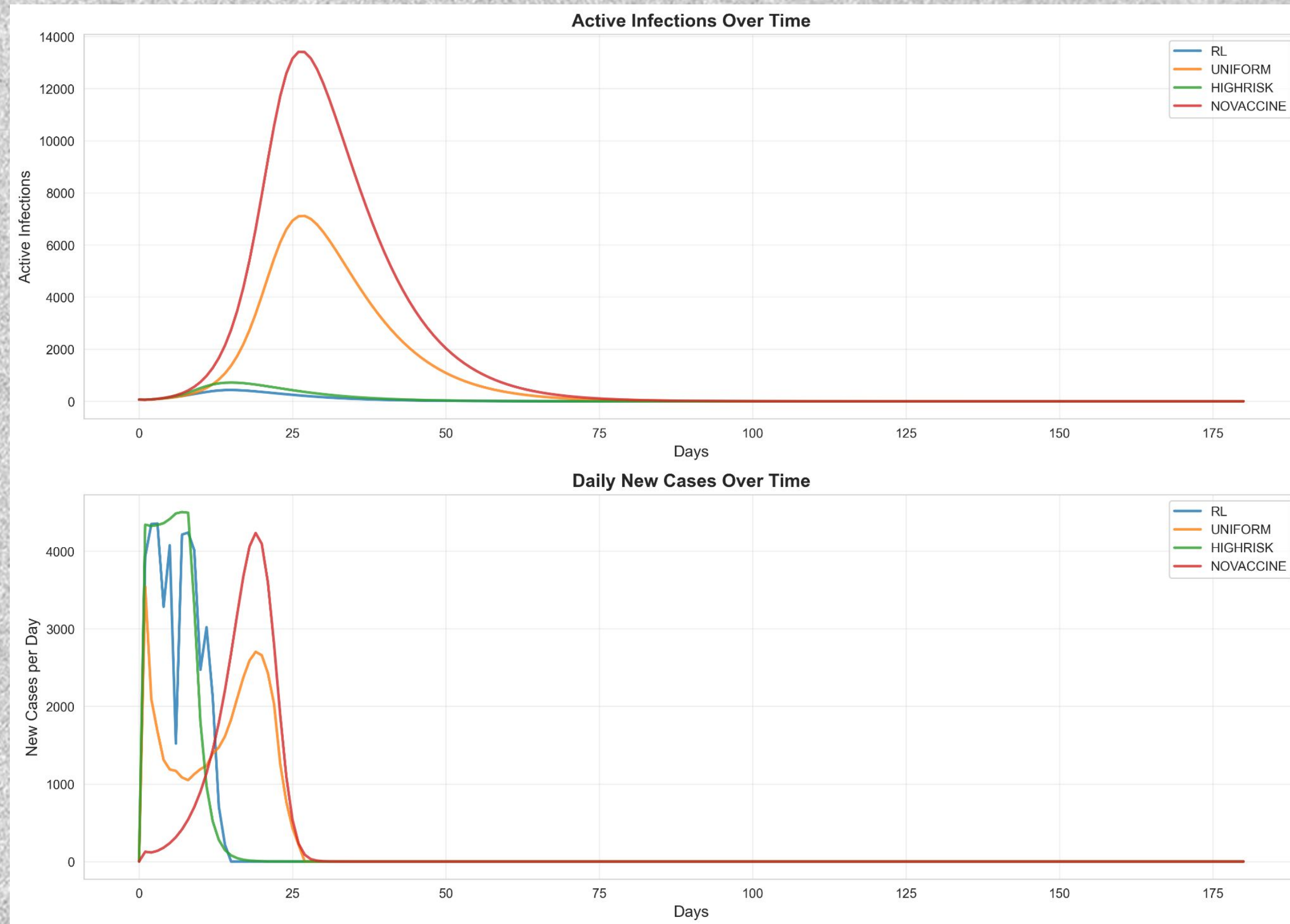
Baseline Simulations Assumptions

- Vaccine hesitancy: 15% baseline (with some random variation)
- Inter-county mobility: 0.2% daily mixing between counties with >10 cases
- Daily vaccine supply: 5,000 doses
- Simulation duration: 180 days
- Immediate vaccine effectiveness (single-dose)

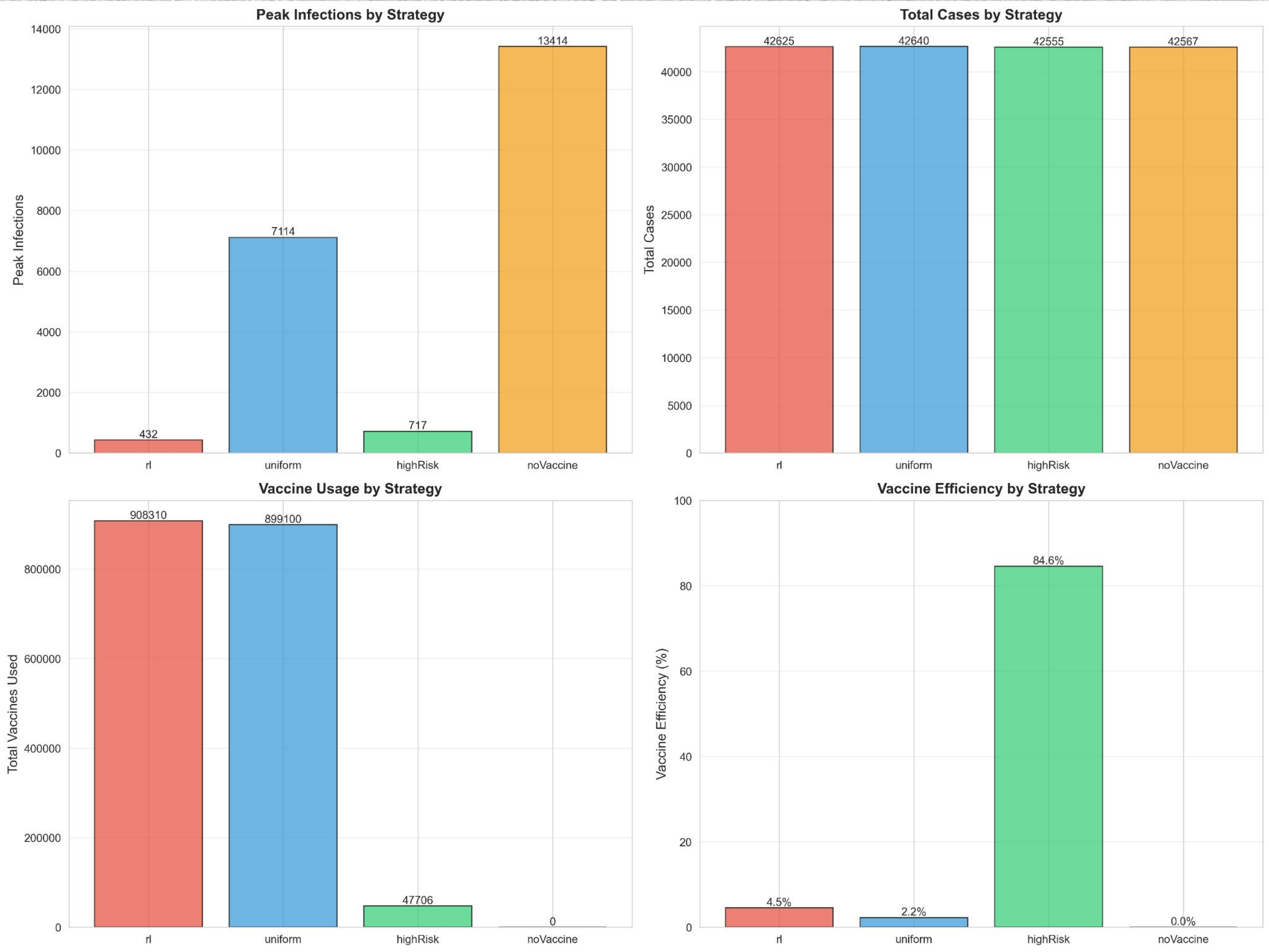
SEIR Dynamics



RL is highly effective... but so is High-Risk



RL uses vaccines inefficiently



**Model
Fitting**

**Spatial Model
Initialization**

**Baseline
Simulations**

**Additional
Experiments**



**Model
Fitting**

**Spatial Model
Initialization**

**Baseline
Simulations**

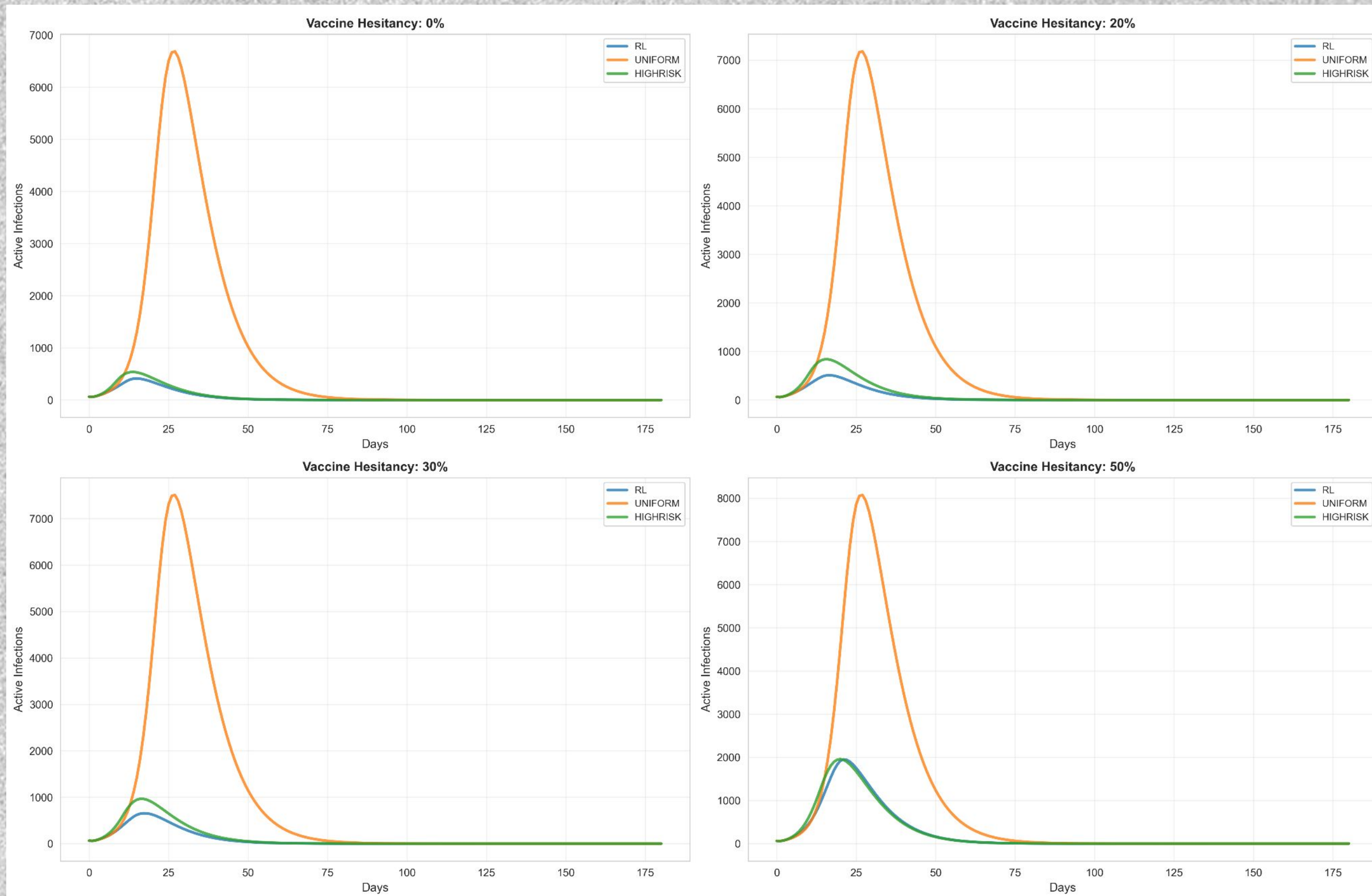
**Additional
Experiments**



Three Experiments

Experiment	Values
Vaccine Hesitancy	0%, 20%, 30%, 50%
Total Vaccine Budget	200k, 450k, 700k, 900k
Daily Vaccine Supply	0, 2500, 5000, 10000

Varying Vaccine Hesitancy

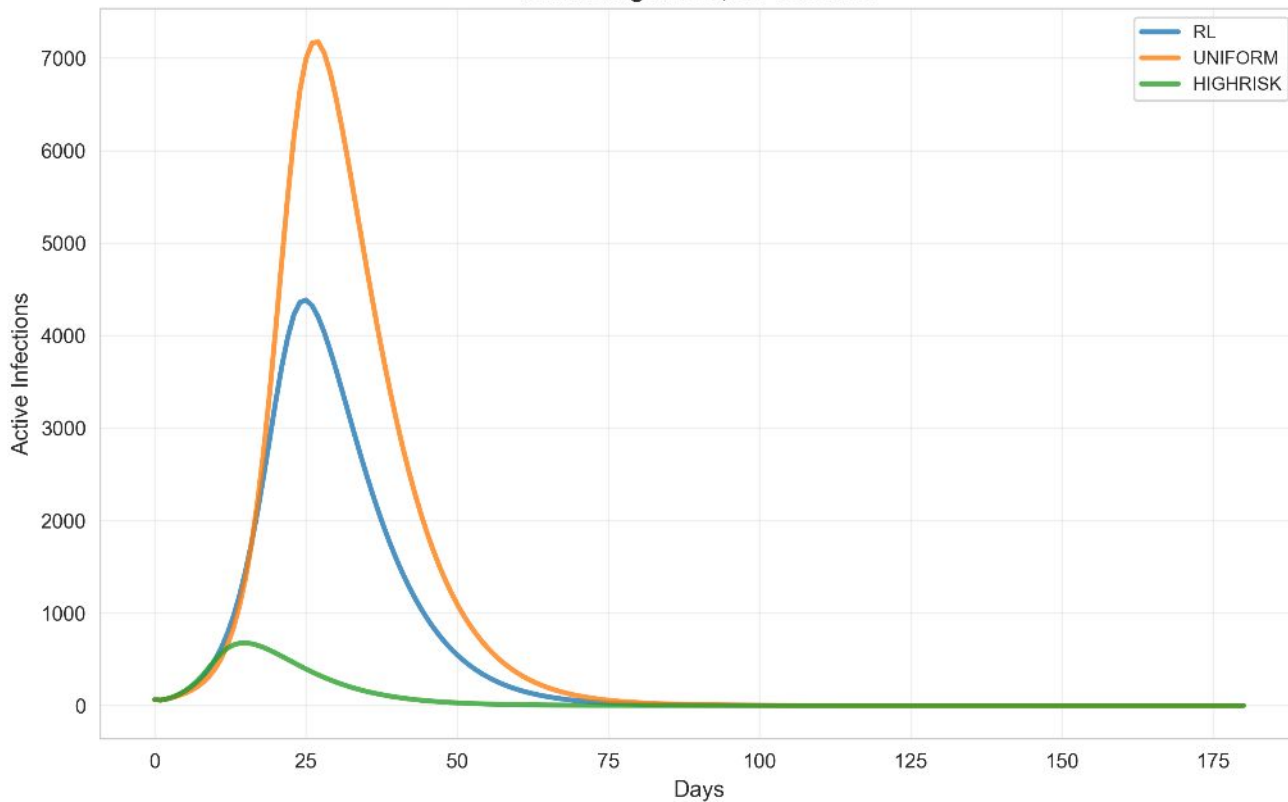


- RL remained lowest across all hesitancy levels
- Hesitancy tipping point that boosts infections
- Higher hesitancy → RL similar to High-Risk

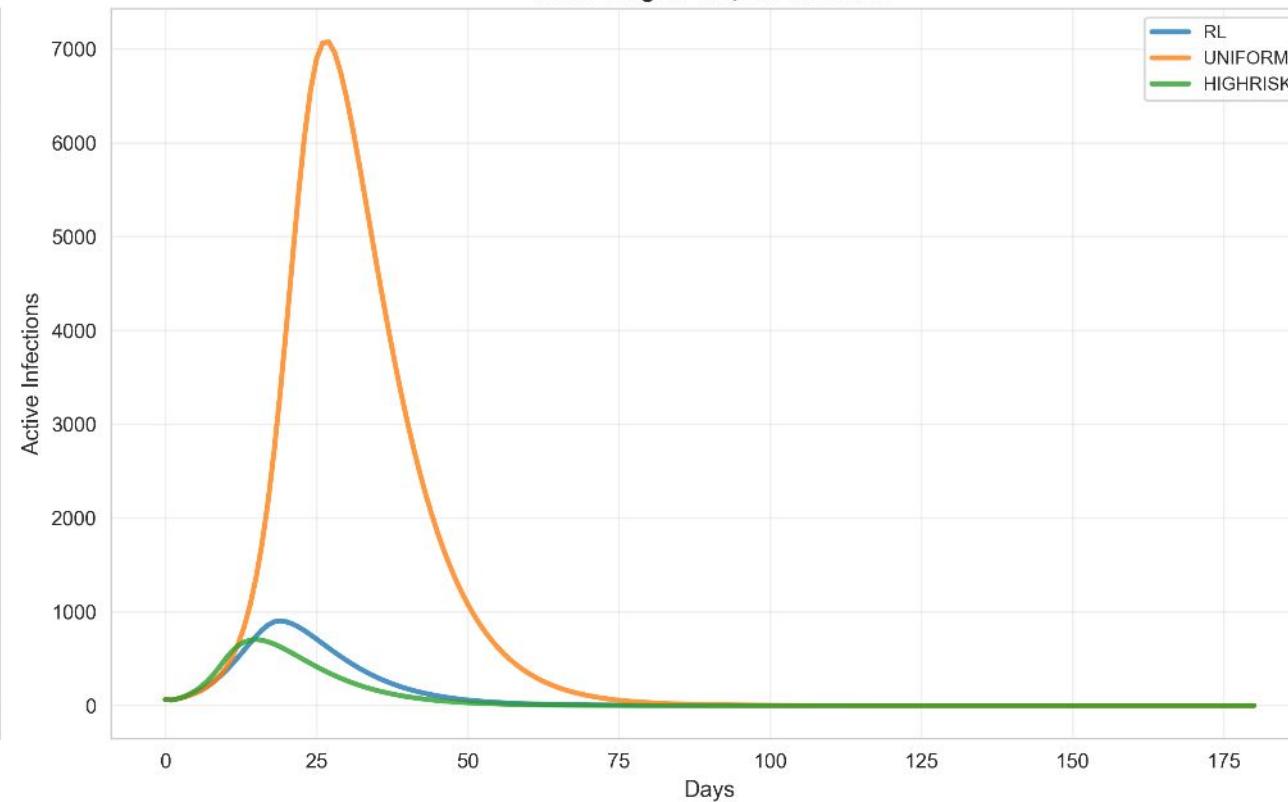
Varying Total Vaccines for Outbreak

- RL worse with limited budget
- Larger budgets → diminishing returns for RL

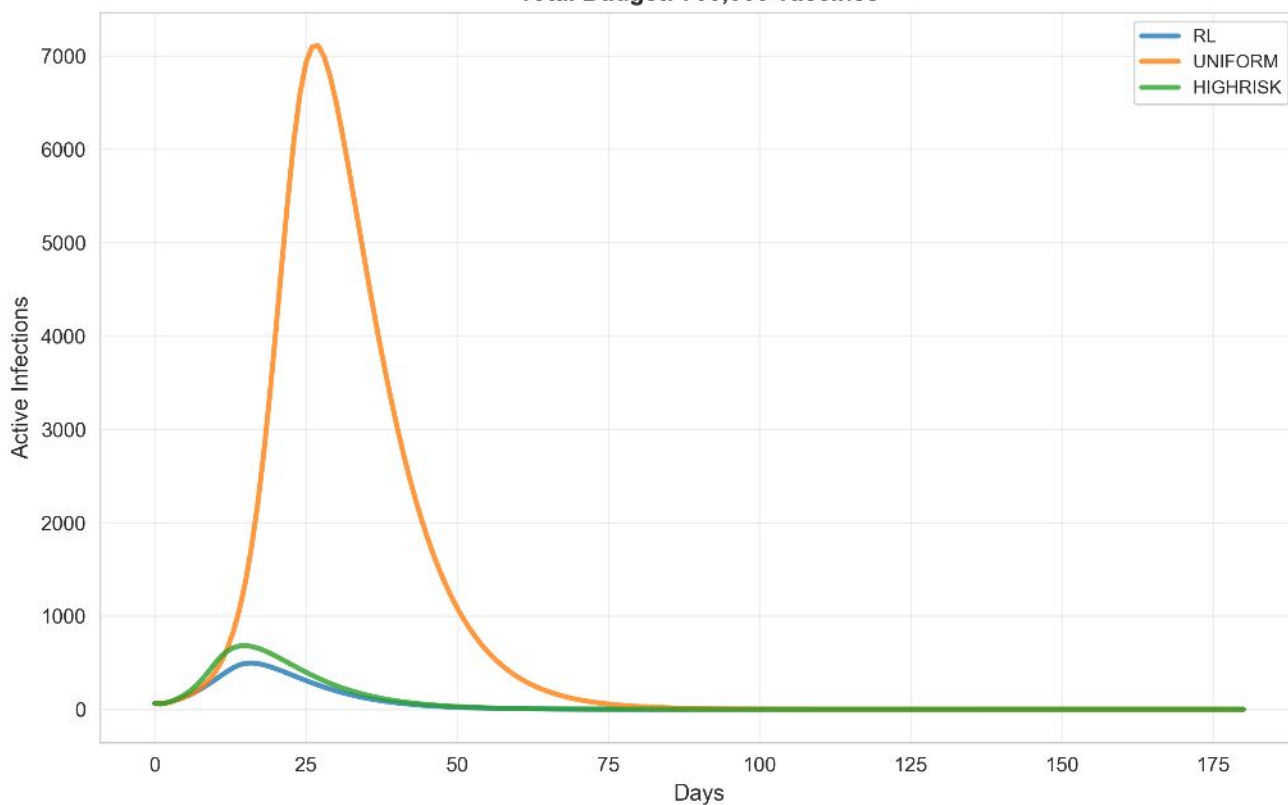
Total Budget: 200,000 vaccines



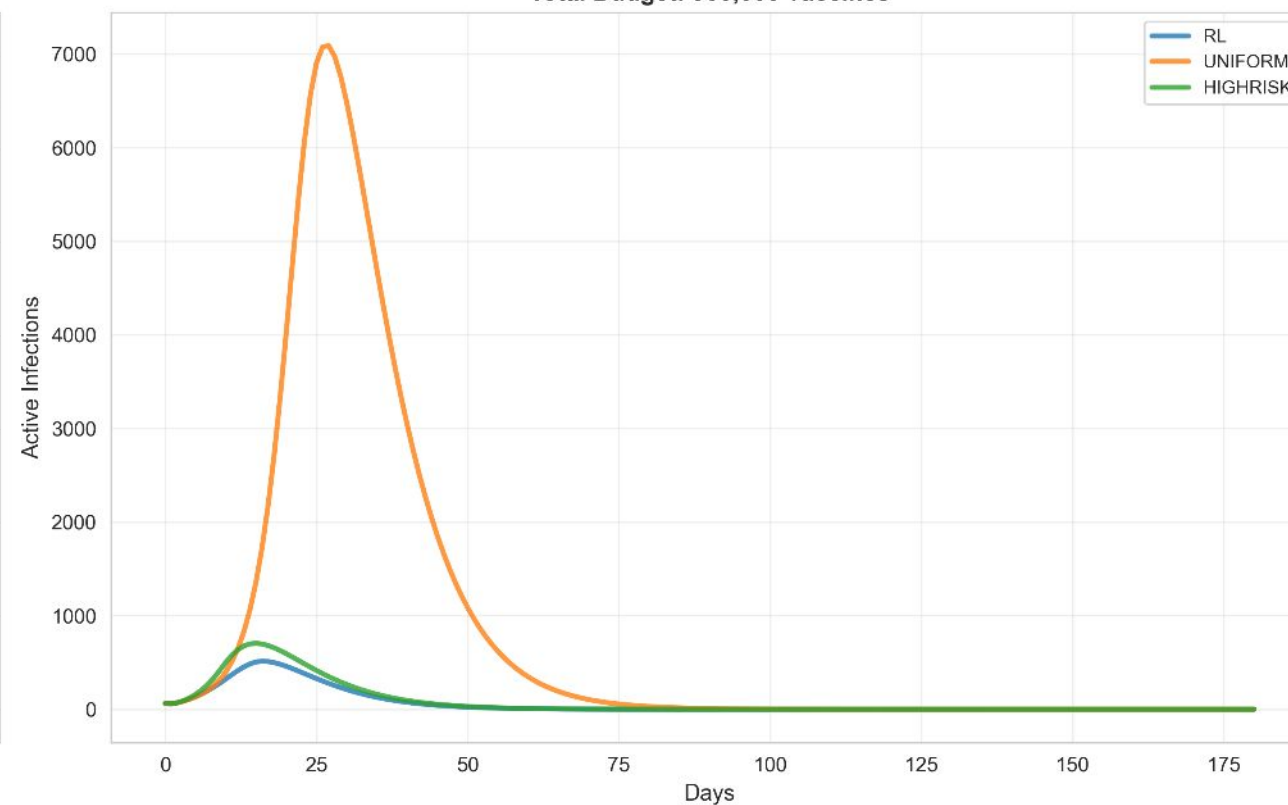
Total Budget: 450,000 vaccines



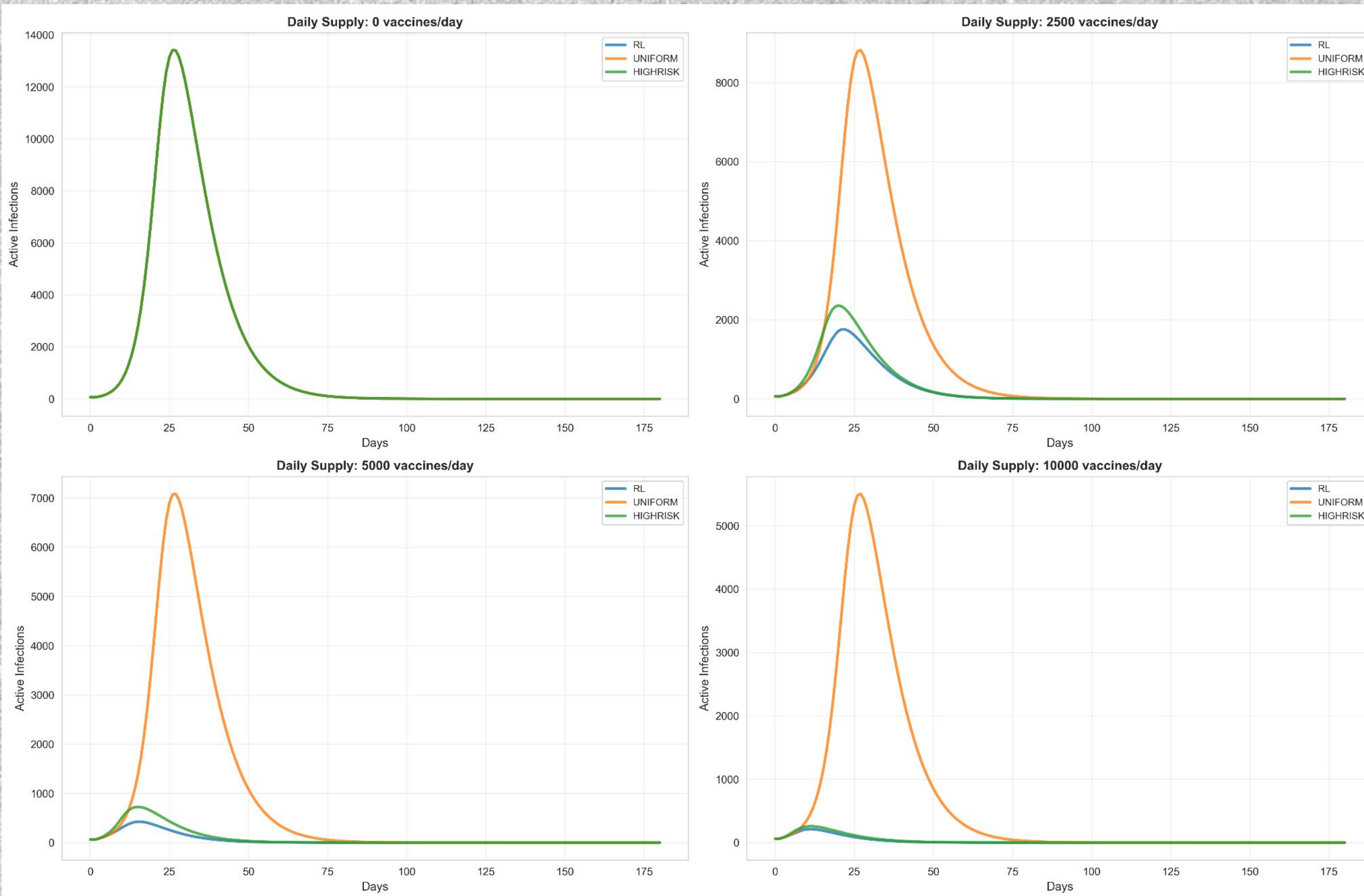
Total Budget: 700,000 vaccines



Total Budget: 900,000 vaccines



Varying Daily Vaccine Supply



- RL remained lowest across all daily vaccine supply levels
- Higher supply → diminishing returns for RL + High-Risk similar effectiveness

Takeaways

- RL does significantly reduce the spread of measles, even when substantial limitations exist, though not efficiently
- This is exacerbated for limits on total vaccines, in which RL performs poorly
 - Highlights the potential for limitations to restrict RL effectiveness
- RL did tend to result in the lowest peak infections, but High-Risk generally performed equally as well
- **AI is not a catch-all solution.** It depends on the priorities of the vaccination program and the limitations at play

Thank You!

References

- Bjørnstad, O. N., Finkenstädt, B. F., & Grenfell, B. T. (2002). Dynamics of Measles Epidemics: Estimating Scaling of Transmission Rates Using a Time Series SIR Model. *Ecological Monographs*, 72(2), 169–184. <https://doi.org/10.2307/3100023>
- Bubar, K. M., Reinholt, K., Kissler, S. M., Lipsitch, M., Cobey, S., Grad, Y. H., & Larremore, D. B. (2021). Model-informed COVID-19 vaccine prioritization strategies by age and serostatus. *Science*, 371(6532), 916–921. <https://doi.org/10.1126/science.abe6959>
- Guerra, F. M., Bolotin, S., Lim, G., Heffernan, J., Deeks, S. L., Li, Y., & Crowcroft, N. S. (2017). The basic reproduction number (R_0) of measles: A systematic review. *The Lancet. Infectious Diseases*, 17(12), e420–e428. [https://doi.org/10.1016/S1473-3099\(17\)30307-9](https://doi.org/10.1016/S1473-3099(17)30307-9)
- Kusmawati, I. S., & Chandra, T. D. (2021). Stability analysis of SIRS epidemic model on measles disease spreading with vaccination and migration. *Journal of Physics: Conference Series*, 1872(1), 012033. <https://doi.org/10.1088/1742-6596/1872/1/012033>
- Ling, L., Mondal, W. U., & Ukkusuri, S. V. (2024). Cooperating Graph Neural Networks With Deep Reinforcement Learning for Vaccine Prioritization. *IEEE Journal of Biomedical and Health Informatics*, 28(8), 4891–4902. <https://doi.org/10.1109/JBHI.2024.3392436>
- Song, B., Wang, X., Sun, P., & Boukerche, A. (2023). Robust COVID-19 vaccination control in a multi-city dynamic transmission network: A novel reinforcement learning-based approach. *Journal of Network and Computer Applications*, 219, 103715. <https://doi.org/10.1016/j.jnca.2023.103715>